

BIST Part 2

- Output Response Analysis

- ◆ Simple ORA

- ◆ LFSR-based ORA

- * Serial : compress one bit at a time

- * Parallel : compress multiple bits at a time

- MISR

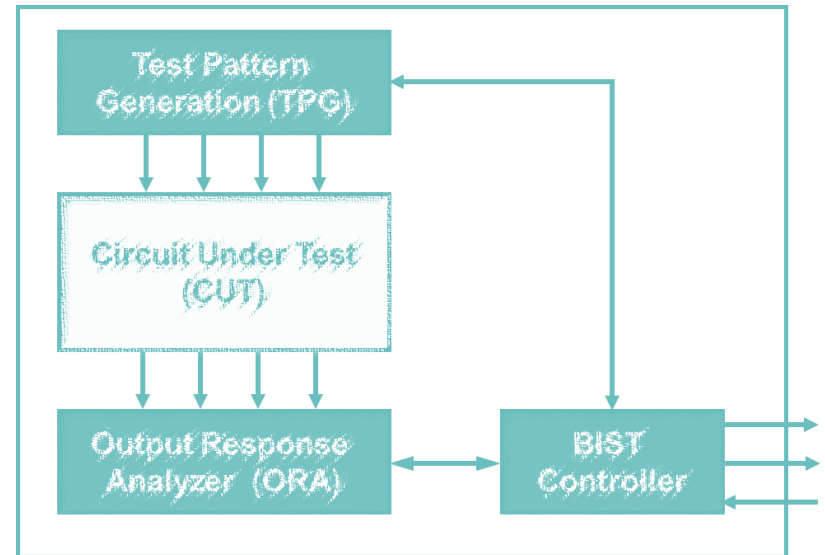
- Aliasing

- Masking

- BIST Architecture

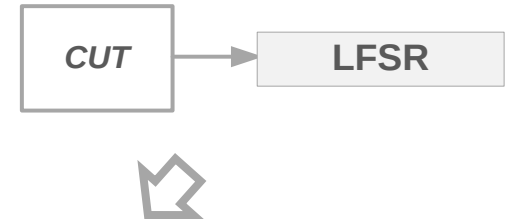
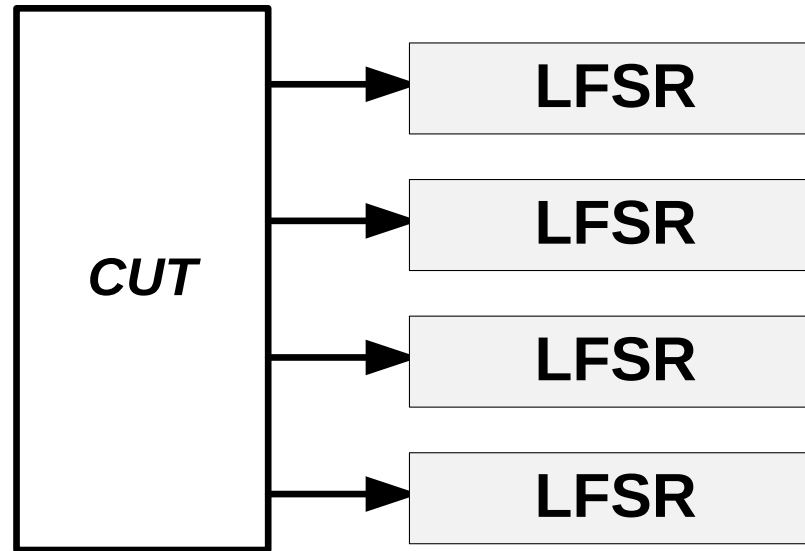
- Issues with BIST

- Conclusions



Parallel ORA

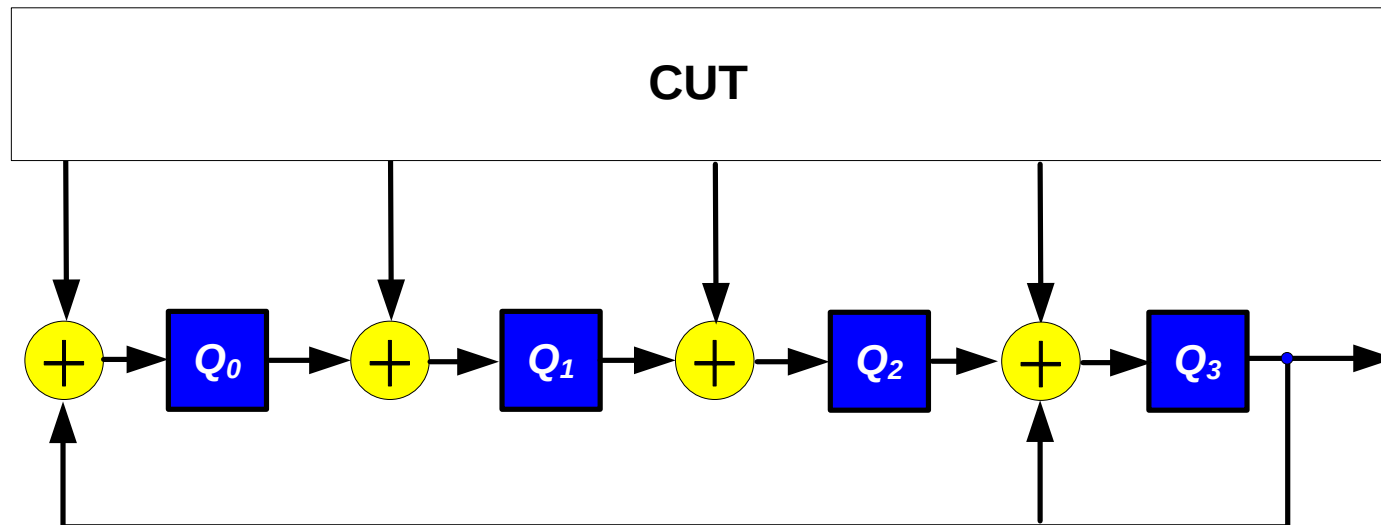
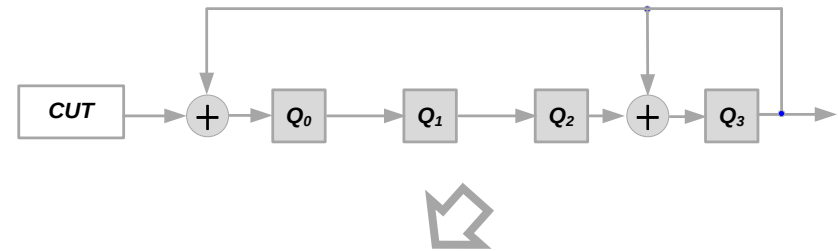
- Serial ORA only compress one CUT output
- How about multiple CUT outputs?
 - ♦ One LFSR for each CUT output?
 - ♦ Too much hardware overhead!



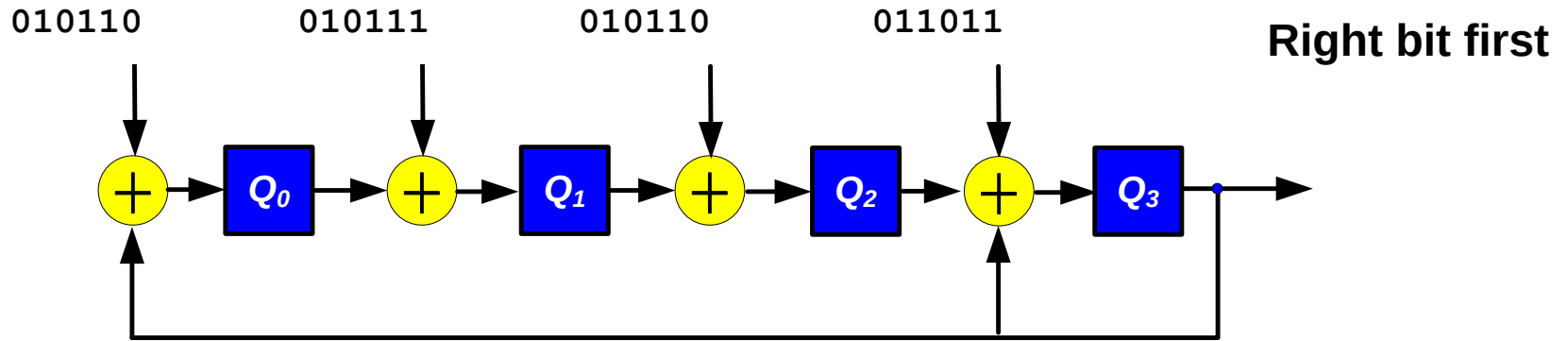
Any Better Idea?

Multiple Input Signature Register, MISR

- **MISR** has similar structure to LFSR, except
 - ♦ Parallel inputs feed XOR between stages
- MISR characteristic polynomial same as LFSR
- Example: MISR with 4 parallel inputs
 - ♦ Modified from type-2 LFSR
 - ♦ $f(x) = x^4 + x^3 + 1$



What is MISR Signature?



cycle	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	0	1	0	1
2	0	1	0	0
3	1	1	0	0
4	0	1	1	1
5	0	1	0	1
6	1	0	1	1

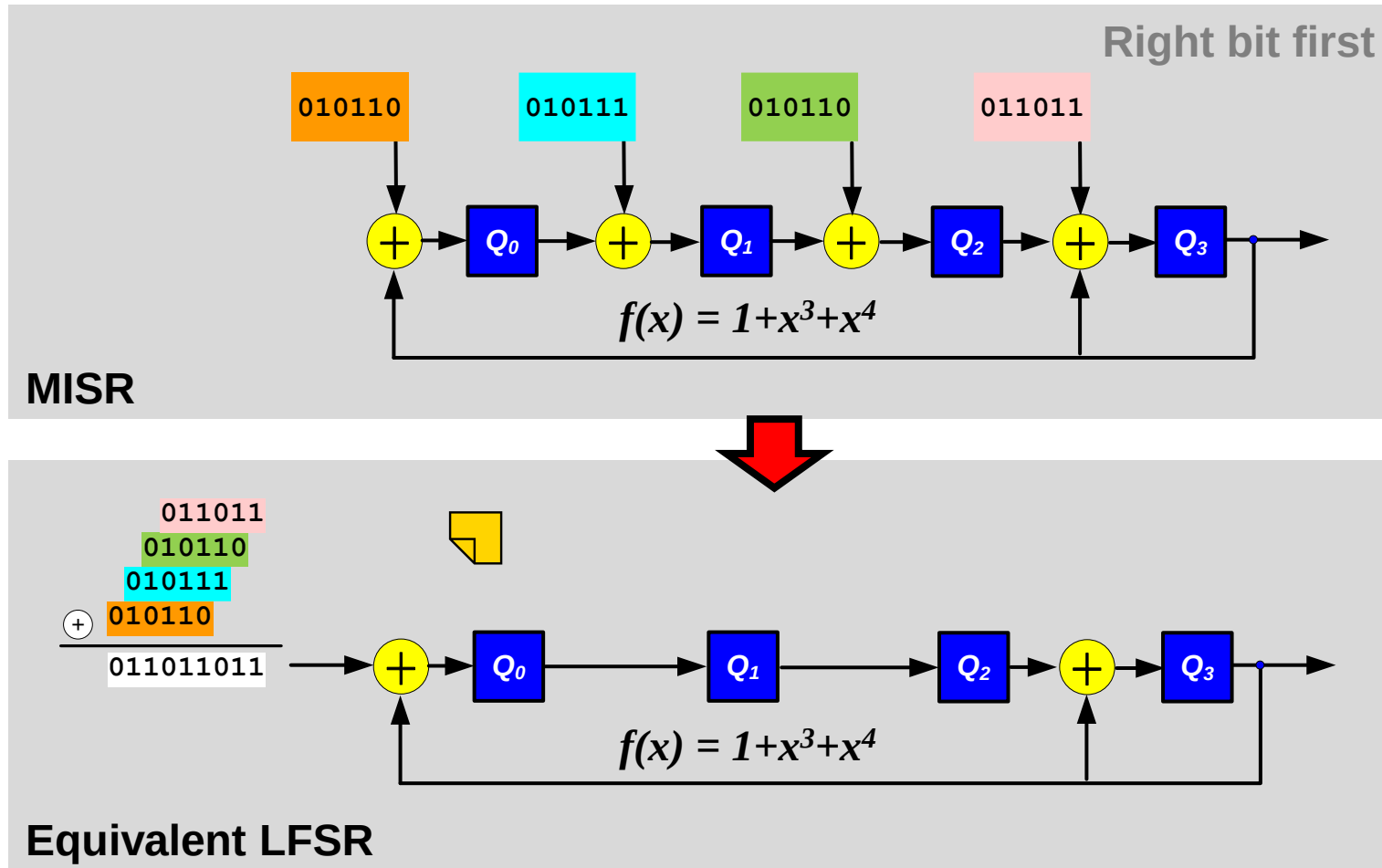
initial state

signature

Too Slow!

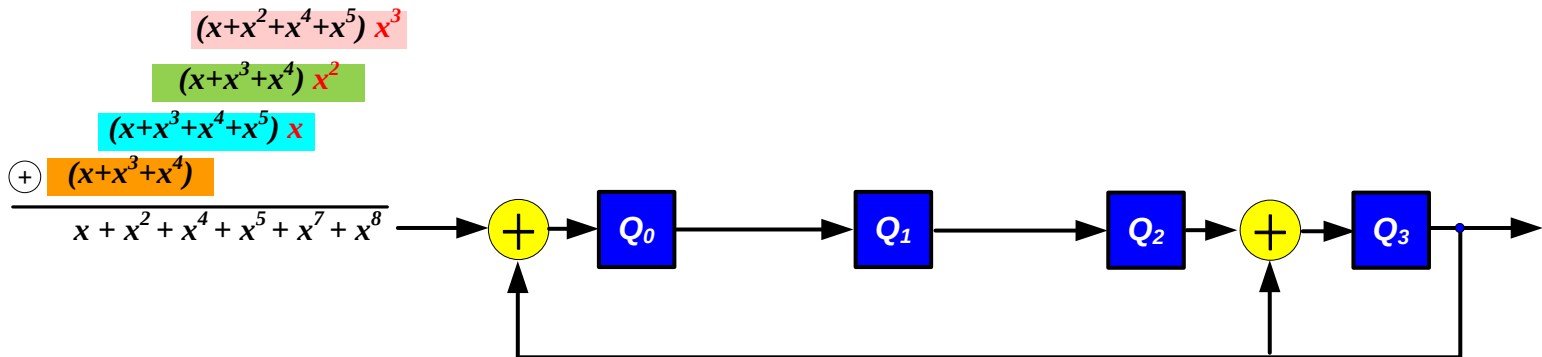
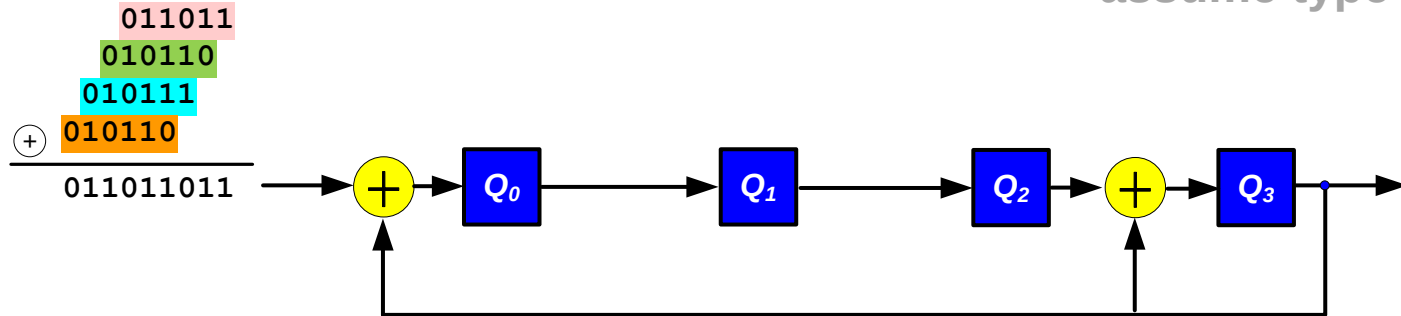
Equivalent LFSR

- Change MISR to *equivalent LFSR*
 - ♦ Just phase shift and add many input bit streams



Signature Analysis Using CRC

assume type-2 LFSR

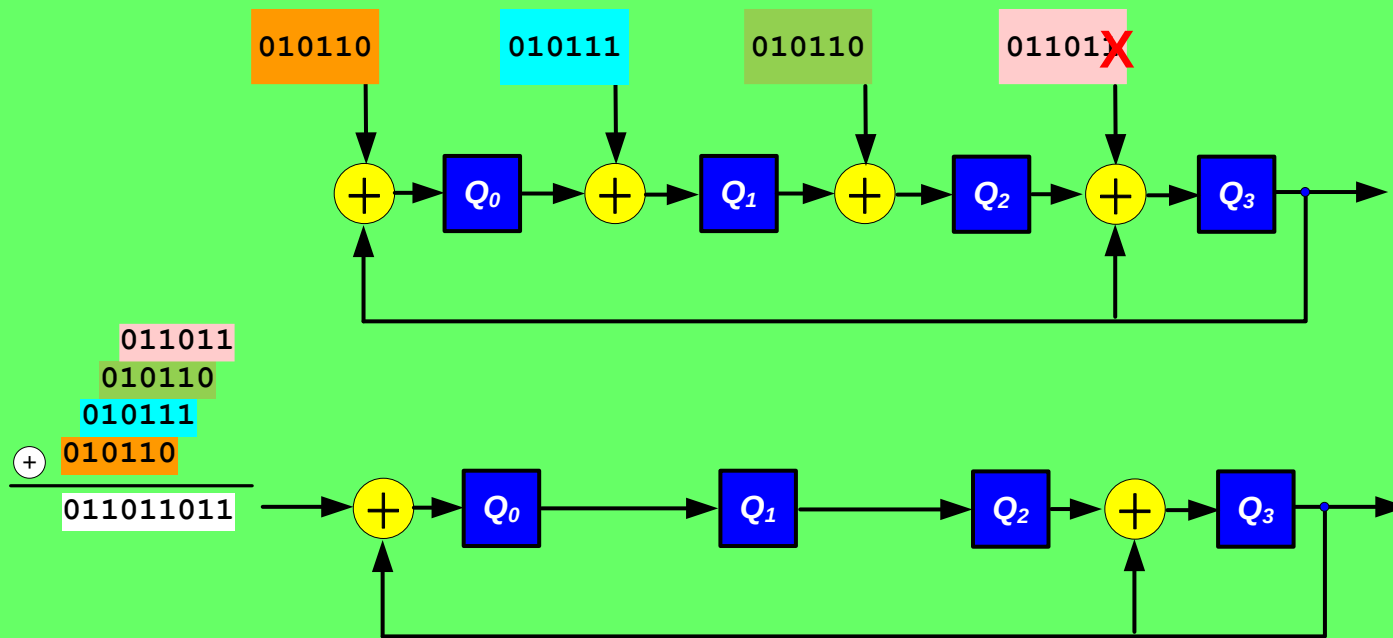


$$(x+x^2+x^4+x^5+x^7+x^8) \div (1+x^3+x^4) = 1+x+x^4 \dots\dots 1+x^2+x^3$$

$$M(x) \quad \div \quad f(x) \quad = \quad Q(x) \quad \dots\dots R(x)$$

Quiz

- Q: Given same example. The first bit of rightmost '1' is flipped to '0'
- 1) Find equivalent LFSR and associated input bit stream
 - 2) What is MISR signature using CRC analysis?



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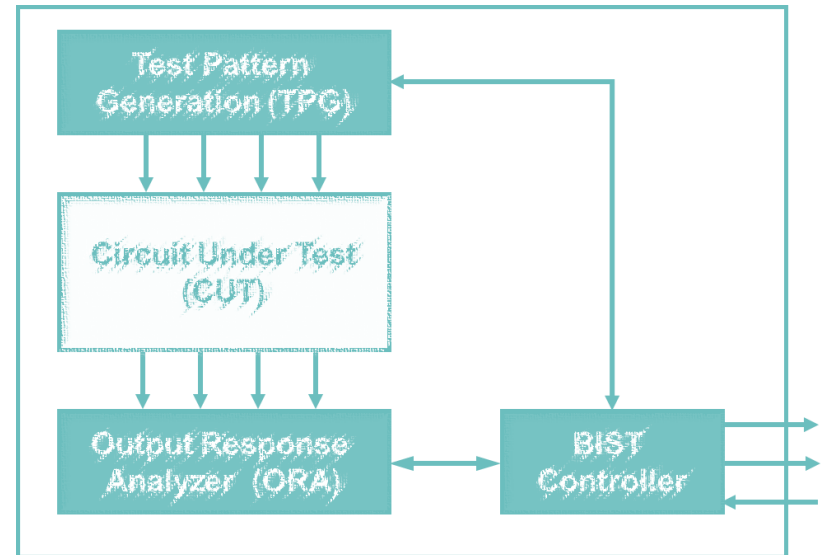
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- Masking

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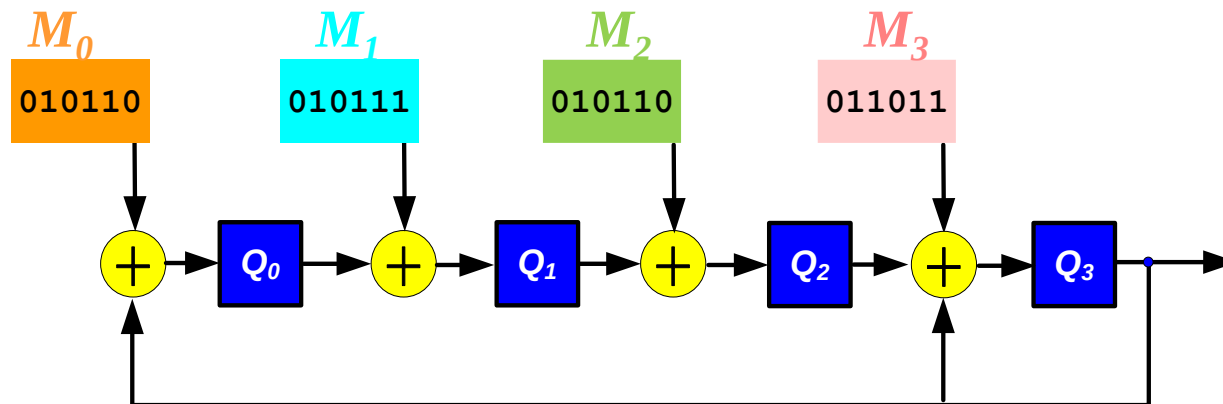
Good Signature (1/2)

- Assume

- ◆ $f(x) = N$ -degree characteristic polynomial, which is primitive
- ◆ Totally N MISR inputs: $n=0,1,2,\dots,N-1$ (from left to right)
- ◆ Totally K cycles of MISR input bit stream: $k=0,1,2,\dots,K-1$
- ◆ $M_n(x)$ = MISR n_{th} input polynomial

- Example

- ◆ $N=4, K=6, f(x) = x^4+x^3+1$
- ◆ $M_0(x)=x+x^3+x^4, M_1(x)=x+x^3+x^4+x^5, M_2(x)=x+x^3+x^4, M_3(x)=x+x^2+x^4+x^5$



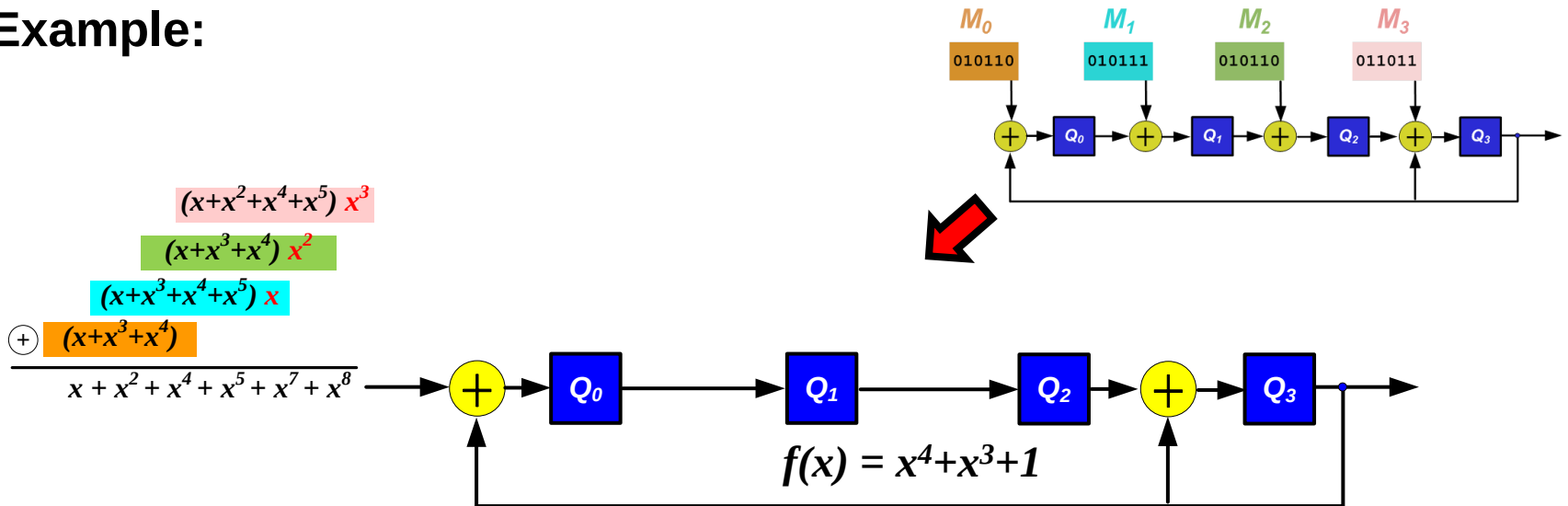
Good Signature (2/2)

- $M_{good}(x)$ = good input bit stream; $R_{good}(x)$ = good signature

$$M_{good}(x) = M_0(x) + x^1 M_1(x) + x^2 M_2(x) \dots + x^{N-1} M_{N-1}(x)$$

$$R_{good}(x) = M_{good}(x) \bmod f(x)$$

- Example:



$$M_{good}(x) = M_0(x) + xM_1(x) + x^2 M_2(x) + x^3 M_3(x) + x^4 M_4(x) = x + x^2 + x^4 + x^5 + x^7 + x^8$$

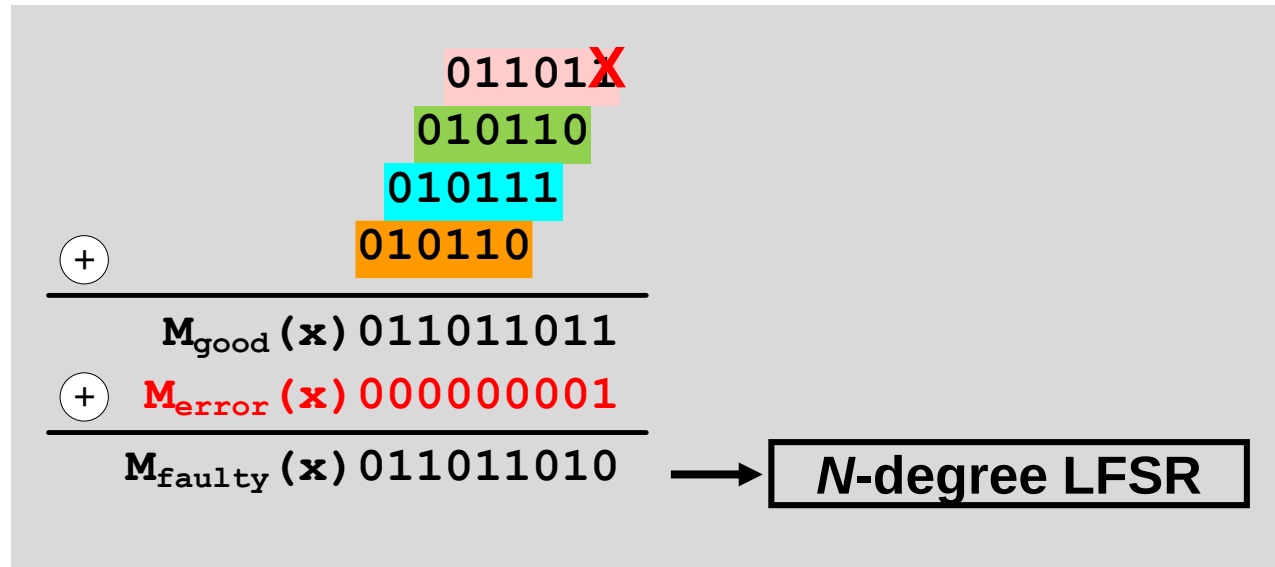
$$R_{good}(x) = M_{good}(x) \bmod f(x) = 1 + x^2 + x^3$$

When Error Occurs

- $M_{\text{faulty}}(x) = M_{\text{good}}(x) + M_{\text{error}}(x)$

$$M_{\text{faulty}}(x) = M_0(x) + x^1 M_1(x) + x^2 M_2(x) \dots + x^{N-1} M_{N-1}(x) + M_{\text{error}}(x)$$

$$R_{\text{faulty}}(x) = M_{\text{faulty}}(x) \bmod f(x)$$



$$M_{\text{faulty}}(x) = x + x^2 + x^4 + x^5 + x^7 + x^8 + \cancel{x^8} = x + x^2 + x^4 + x^5 + x^7$$

$$R_{\text{faulty}}(x) = M_{\text{faulty}}(x) \bmod f(x) = 1 + x \quad \text{No aliasing.}$$

PAL=?

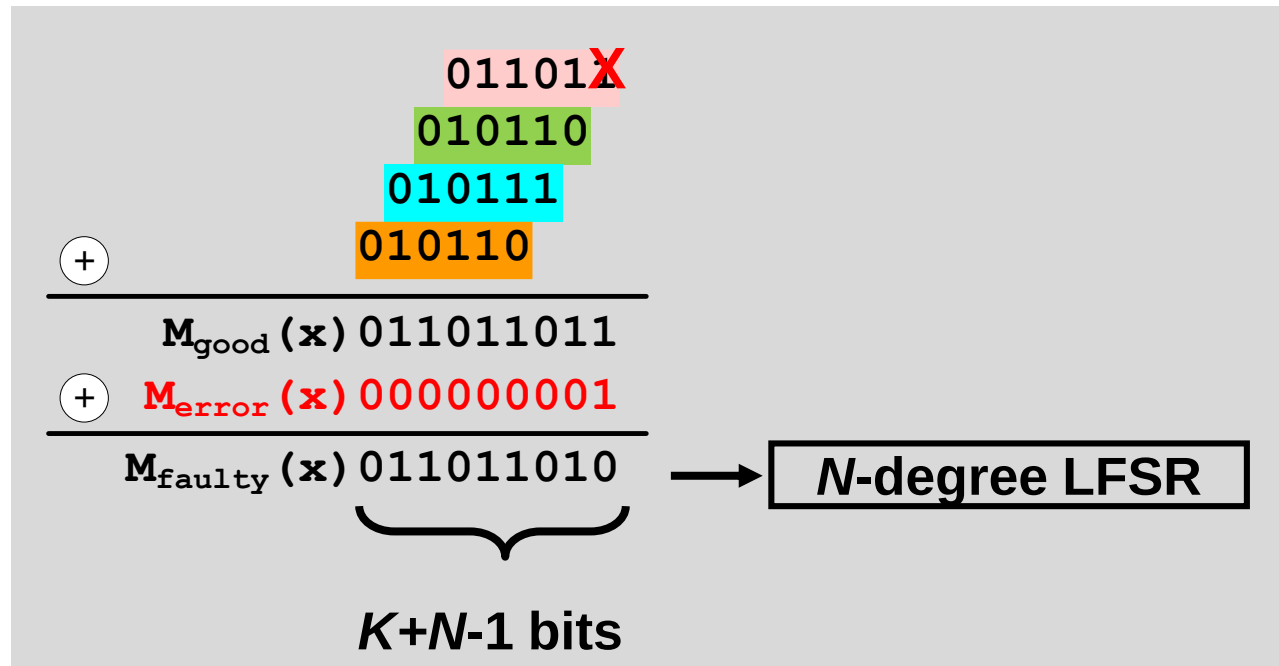
- Same analysis as LFSR

- ♦ $M_{\text{faulty}}(x) = M_{\text{good}}(x) + M_{\text{error}}(x)$
- ♦ Bit stream length $m = K + N - 1$

$$PAL = \frac{2^{m-N} - 1}{2^m - 1} = \frac{2^{K-1} - 1}{2^{N+K-1} - 1} \approx 2^{-N}$$

- Example

- ♦ $N=4, K=6$
- ♦ $m=9$
- ♦ $PAL \approx 2^{-4}$

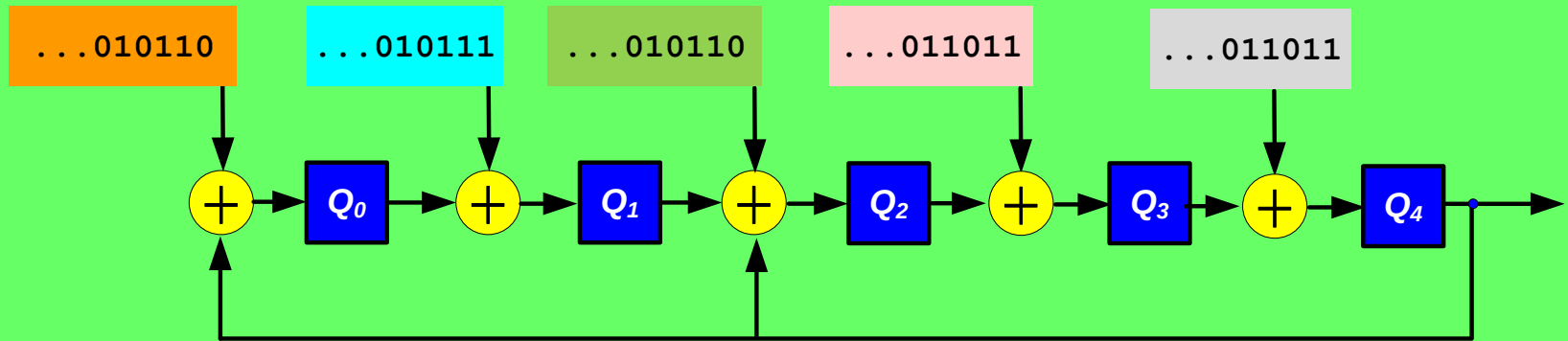


$PAL = 2^{-N}$. Same as LFSR!

Quiz

Q: We have 5 inputs, each 100 bits long. 5-degree MISR (PP= $1+x^2+x^5$)
What is PAL of this MISR?

A:



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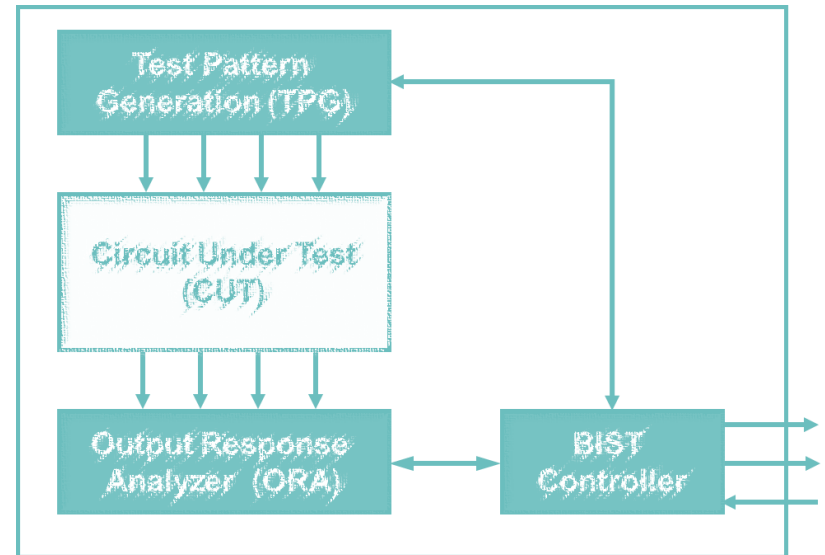
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What If Two Errors?

- Two cases:

different columns

$$\begin{array}{r}
 \begin{array}{r}
 011011 \\
 01011\cancel{0} \\
 0101\cancel{1}1 \\
 010110
 \end{array} \\
 \oplus \\
 \hline
 M_{\text{good}}(x) \ 011011011 \\
 \oplus \ M_{\text{error}}(x) \ 00000\color{red}{1}0\color{red}{1}0 \\
 \hline
 M_{\text{faulty}}(x) \ 01101\color{red}{0}0\color{red}{0}1
 \end{array}$$

This is modeled by PAL

same column

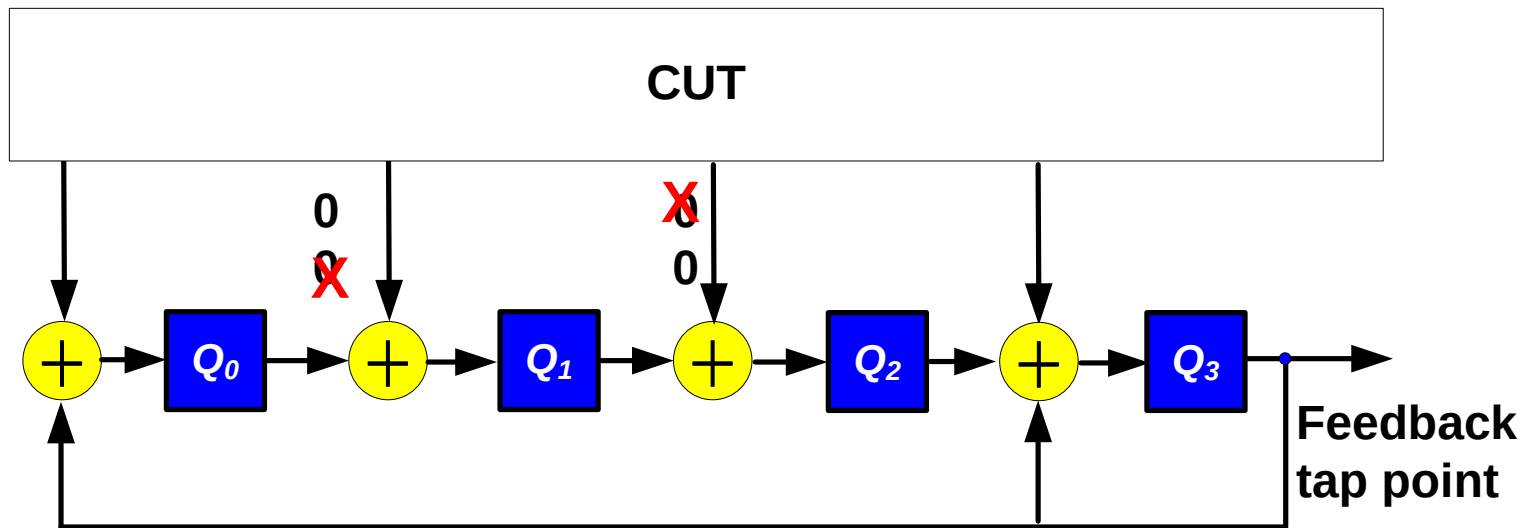
$$\begin{array}{r}
 \begin{array}{r}
 011011 \\
 010\cancel{1}0 \\
 0101\cancel{1}1 \\
 010110
 \end{array} \\
 \oplus \\
 \hline
 M_{\text{good}}(x) \ 011011011 \\
 \oplus \ M_{\text{error}}(x) \ 00000\color{red}{0}000 \\
 \hline
 M_{\text{faulty}}(x) \ 011011011
 \end{array}$$

Error output → Good signature
But this is NOT aliasing
What is wrong?

Masking

- **Masking** (aka. **error cancellation**)

- ♦ One error bit cancels another error bits
- ♦ **Before** reaching MISR feedback tap points



Masking $\neq$ Aliasing

More Careful Analysis

- Actually, error can happen anywhere in the middle
 - ♦ Error bits can get cancelled before summation

$$M_{\text{faulty}}(x) = M_0(x) + M_{\text{error}0}(x) + x^1 M_1(x) + M_{\text{error}1}(x) + x^2 M_2(x) + M_{\text{error}2}(x) + \dots$$

$$\begin{array}{r}
 \phantom{M_{\text{error}0}(x)} 011011 \\
 M_{\text{error}0}(x) 000000 \\
 \phantom{M_{\text{error}0}(x)} 010110 \\
 M_{\text{error}1}(x) 000100 \\
 \phantom{M_{\text{error}0}(x)} 010111 \\
 M_{\text{error}2}(x) 000010 \\
 \phantom{M_{\text{error}0}(x)} 010110 \\
 \oplus \phantom{M_{\text{error}0}(x)} M_{\text{error}3}(x) 000000 \\
 \hline
 M_{\text{faulty}}(x) 011011011
 \end{array}$$

Probability of Masking (Bardell 87)

- Probability of masking = Probability of even ones in same column

$$\begin{array}{r}
 M_{\text{error}0}(x) \quad 000000 \\
 M_{\text{error}1}(x) \quad 000100 \\
 M_{\text{error}2}(x) \quad 000010 \\
 \oplus \quad M_{\text{error}3}(x) \quad 000000 \\
 \hline
 M_{\text{error}}(x) \quad 0000000000
 \end{array}$$

- Assume all errors are equally likely, then

$$\Pr(\text{masking}) \approx 2^{1-N-K} \ll 2^{-N} = \text{PAL}$$

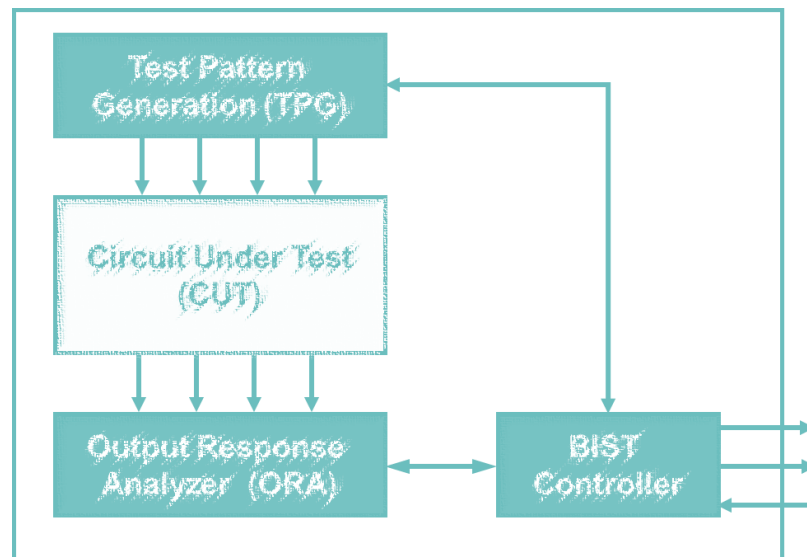
- Conclusion: Prob. of masking is much smaller than PAL

Aliasing and Masking

	Aliasing	Masking
Reason of happening	Error bits propagate through feedback path and cancel out with errors in later cycles	Error bit shifted down the MISR and cancelled by another error bit, before reaching feedback tap points
Probability	2^{-N}	2^{1-N-K} Smaller
Happens in	Both LFSR and MISR	Only MISR
Polynomial	Dependent 	Independent 

Summary

- **MISR** (Multiple input signature register)
 - ◆ Similar to LFSR but multiple inputs
 - ◆ **Most popular parallel ORA**
 - * Small area, low PAL
- How to analyze MISR?
 - ◆ Convert to **equivalent LFSR**
- Aliasing
 - ◆ **$PAL = 2^{-N}$** , same as LFSR
- Masking
 - ◆ **Prob. much smaller than PAL**



FFT

- Q: Why is aliasing PAL polynomial-dependent?
 - ♦ What if non-primitive polynomial 

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